

Written Assignment 6

Grant Talbert

Exercise 1. Let $B = \{(1, 0), (0, 1)\}$ and $B' = \{(1, 1), (1, 2)\}$ be bases for $\mathbb{R}^2$.

1. Find the matrix A for $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (x - 2y, x + 4y)$, relative to the basis B .
2. Find the transition matrix from $\mathcal{B}$ to $\mathcal{B}'$.
3. Find the matrix A' of T relative to the basis B' .
4. Find $[T(\vec{v})]_{B'}$ when $[\vec{v}]_{B'} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

Solution. Trivially, $T(1, 0) = (1, 1)$ and $T(0, 1) = (-2, 4)$. Define

$$A := \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$$

and notice that T has the representation $T : \mathbf{x} \mapsto A\mathbf{x}$.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \xrightarrow{rref} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 1 \end{bmatrix}$$

Thus, the matrix

$$P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

gives a transition from the basis $\mathcal{B}$ to $\mathcal{B}'$. Also notice that its inverse will be

$$P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

which gives a transition from $\mathcal{B}'$ to $\mathcal{B}$. Thus, we also know that A' is given as

$$A' = P^{-1}AP = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -7 & -15 \\ 6 & 12 \end{bmatrix}$$

Thus we have

$$[T(\mathbf{v})]_{B'} = \begin{bmatrix} -7 & -15 \\ 6 & 12 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \end{bmatrix} = [\mathbf{v}]_{B'}$$

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Exercise 2. Prove that if A and B are similar matrices, then $|A| = |B|$.

Proof. Let the relation $\sim$ denote similar matrices and let $A, B \in M_{n \times n}(\mathbb{F})$ have $A \sim B$. Thus,

$$\exists P \in \text{GL}_n(\mathbb{F}) : A = P^{-1}BP$$

by definition of similar matrices. Define P to be such a matrix for which the above condition holds, and observe that

$$A = P^{-1}BP \implies \det(A) = \det(P^{-1}BP)$$

From repeated applications of one of the more important theorems in Chapter 3,

$$\det(P^{-1}BP) = \det(P^{-1}) \det(B) \det(P)$$

From another one of the theorems, we know that for any P ,

$$\det(P^{-1}) = \frac{1}{\det(P)}$$

Therefore, we have

$$\begin{aligned}\det(A) &= \det(P^{-1}BP) \\ &= \det(P^{-1})\det(B)\det(P) \\ &= \frac{\det(P)}{\det(P)}\det(B) \\ &= \det(B)\end{aligned}$$

Since we define P such that P^{-1} exists, $\det(P) \neq 0$, and the above statements hold. Thus, $\det(A) = \det(B)$, and therefore

$$A \sim B \implies \det(A) = \det(B)$$

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Exercise 3. Prove that if matrix A is diagonalizable, then A^T is diagonalizable.

Proof. For the sake of simplicity, let

$$\mathfrak{D}_n(\mathbb{F}) := \{A \in M_{n \times n}(\mathbb{F}) \mid A \text{ is diagonal}\}$$

Let $A \in M_{n \times n}(\mathbb{C})$. We say that A is diagonalizable if there exists some matrix $P \in \text{GL}_n(\mathbb{C})$ such that

$$P^{-1}AP \in \mathfrak{D}_n(\mathbb{C})$$

where D is some diagonal matrix. In other words, A is similar to some diagonal matrix D . Let P be a matrix where this is true, and define $D \in \mathfrak{D}_n(\mathbb{C})$ as

$$D := P^{-1}AP$$

It follows directly that

$$\begin{aligned}A &= PDP^{-1} \\ \implies A^T &= (PDP^{-1})^T\end{aligned}$$

Recall that for matrices X, Y , $(XY)^T = Y^T X^T$. Thus,

$$\begin{aligned}A^T &= (PDP^{-1})^T \\ &= (P(DP^{-1}))^T \\ &= (DP^{-1})^T P^T \\ &= (P^{-1})^T D^T P^T\end{aligned}$$

Recall that for some $X \in \text{GL}_n(\mathbb{F})$, $(X^{-1})^T = (X^T)^{-1}$. Thus,

$$A^T = (P^{-1})^T D^T P^T = (P^T)^{-1} D^T P^T$$

Trivially, the transpose of a diagonal matrix is diagonal, so $D^T \in \mathfrak{D}_n(\mathbb{C})$. Furthermore, since P^T and $(P^T)^{-1}$ are inverses, they must be invertible matrices. Define $C := P^T$, and as seen previously, $C \in \text{GL}_n(\mathbb{C})$. We have

$$A^T = C^{-1}D^TC$$

By definition of similar matrices, A^T is similar to D^T , and therefore A^T is diagonalizable. ■

Exercise 4. Let $A = \begin{bmatrix} 2 & -2 & 3 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}$.

1. Find the characteristic polynomial of A .
2. Find the eigenvalues of A .
3. Find the corresponding eigenvectors for the eigenvalues found in part (b).
4. Is the matrix A diagonalizable? Why or why not?

Solution. Let λ be an eigenvalue for A with eigenvector $\mathbf{x}$.

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 2 & 2 & -3 \\ 0 & \lambda - 3 & 2 \\ 0 & 1 & \lambda - 2 \end{vmatrix} = (\lambda - 2)((\lambda - 3)(\lambda - 2) - 2) = 0$$

$$\implies (\lambda - 2)(\lambda^2 - 5\lambda + 4) = \lambda^3 - 7\lambda^2 + 14\lambda + 8 = 0$$

is the characteristic polynomial of A , which has three roots in $\mathbb{C}$. Starting from the form $0 = (\lambda - 2)(\lambda^2 - 5\lambda + 4)$ and factoring fully, we obtain

$$0 = (\lambda - 2)(\lambda - 4)(\lambda - 1)$$

Since the characteristic polynomial factors into three distinct linear factors over $\mathbb{Q}$, we have three real-valued eigenvalues $\lambda \in \{1, 2, 4\}$. Take $\mathbf{x} = (x, y, z)$ and let

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

For $\lambda = 1$, it follows that

$$\begin{bmatrix} -1 & 2 & -3 \\ 0 & -2 & 2 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We give the representation

$$\begin{bmatrix} -1 & 2 & -3 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \xrightarrow{rref} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus, a basis for the eigenspace of $\lambda = 1$ is $E_1 = \{(-1, 1, 1)\}$. Similarly, for $\lambda = 2$,

$$\begin{bmatrix} 0 & 2 & -3 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{rref} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

implies that $E_2 = \{(1, 0, 0)\}$. Finally, for $\lambda = 4$,

$$\begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \xrightarrow{rref} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

only the trivial solution exists. Thus, $E_3 = \emptyset$ since the eigenspace is the trivial vector space. Notice that every eigenvector of A is linearly independent of the others. Therefore, A is diagonalizable. ■

Exercise 5. Let $T : P_1 \rightarrow P_1$ be a linear transformation such that

$$T(a + bx) = a + (a + 2b)x$$

If possible, find a basis B for P_1 such that the matrix for T relative for B is diagonal.

Solution. For convenience, we represent P_1 with $\mathbb{R}^2$, since they are isomorphic. We give the matrix A for which $T(\mathbf{x}) = A\mathbf{x}$:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$$

Trivially, this matrix has eigenvalues 1 and 2. This matrix will be diagonalizable. Now consider the reduced row-echelon form of the matrices $\lambda I - A$.

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & -1 & 0 \end{bmatrix} \xrightarrow{rref} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, the eigenvector $(1, -1)$ forms a basis for the eigenspace given by $\lambda = 1$.

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \xrightarrow{rref} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, the eigenvector $(0, 1)$ forms a basis for the eigenspace given by $\lambda = 2$. Taking these eigenvectors, define

$$P := \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

which has inverse

$$P^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Consider the matrix

$$D := P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

which is diagonal. Thus, D is a diagonal matrix giving the transformation T on the basis induced by P^{-1} , which can be realized via the equivalent statement

$$A = PDP^{-1}$$

This statement makes it obvious that D represents the map T on the basis induced by P^{-1} . To solve the problem, we must identify this basis and give its representation in P_1 . First, let $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{B}$, and take

$$\mathbf{v}_1 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$\mathbf{v}_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Converting back to P_1 we say that the basis set

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ x \end{bmatrix}, \begin{bmatrix} 0 \\ x \end{bmatrix} \right\}$$

gives the transformation T as a diagonal matrix. ■